

File Comparison Report

Generated by t0900 on 20-May-2026

Files		
File name	Left File gilbreathOLOP.m	Right File gilbreathOLOP_thesis.m
File path	C:\Users\t0900\Documents\MATLAB\Volkan	C:\Users\t0900\Documents\MATLAB\Volkan
Last modified	20-May-2026 11:22:55	20-May-2026 13:38:41

Environment
MATLAB 26.1 (R2026a)

Comparison Results

+ Insertion - Deletion ≠ Modification

1	%***** ***** olop.m ***** *****	1	%***** ***** olop.m ***** *****
2	% Open-loop Onset Point (OLOP) analysis for the aircraft system with a	2	%This M-file is used for application of the Open-loop Onset Point (OLOP) to an aircraft system
3	% rate-limiting element.	3	%with a rate-limiting element in it . Two simulink models are needed for the analysis. The first is
4	%		
5	% Aircraft model: AIAA-95-3204-CP (Duda / Hanke / Gilbreath).		
6	%		
7	%=====		
8	% CORRECTIONS APPLIED TO THE ORIGINAL gilbreathOLOP.m by CLAUDE AI		
9	%=====		
10	%		
11	% FIX 1 - LINMOD removed (the original Simulink models nt33v2Dlin.mdl and		
12	% nt33v2dolop.mdl are not available). The two linmod calls are		
13	% replaced with analytical closed-loop transfer functions:		
14	%		
15	% Yc = ss(A ,B ,C (4,:) ,D (4,:)) -> K*Gc*Gac / (1 + Gc*Gac)		
16	% P_RLE = ss(A ,B ,C (6,:) ,D (6,:)) -> K*Gc / (1 + Gc*Gac)		
17	% OLOP = ss(AA,BB,-CC,-DD) -> Pilot * Yc		

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18 %
19 %     The "-CC,-DD" sign flip i
n the original was a linmod
20 %     feedback-sign convention a
djustment; the analytical form needs
21 %     no flip because Yc alread
y carries the correct sign.
22 %
23 % FIX 2 - lin_abcd.m inlined. The ai
rcraft model constants K, Gc, Gac
24 %     are defined directly belo
w this header instead of being loaded
25 %     from an external script. T
he script is now self-contained.
26 %
27 % FIX 3 - ngridneal embedded as a lo
cal function at the bottom of this
28 %     file. The original referen
ced an ngridneal.m that ships with
29 %     the Gilbreath OLOP packag
e but is not in standard MATLAB.
30 %     The local copy draws the s
tandard Nichols grid plus the +3 dB
31 %     M-contour (Neal-Smith Leve
l-1/2 closed-loop resonance bound).
32 %
33 % FIX 4 - Margin-warning suppressio
n bracket around the Kg pilot-gain
34 %     sweep. margin(Kg(i)*Yc) em
its "closed-loop unstable" warnings
35 %     at high gains; the bracke
t silences only that specific warning
36 %     and restores it after th
e sweep.
37 %
38 % FIX 5 - nichols(OLOP,w) output squ
eeze. Like bode, nichols returns
39 %     1xN×Nw 3-D arrays for SIS
O inputs. magN and phN are squeezed
40 %     to row vectors so phN(k),
phN(iLo:iHi) etc. work in 1-D.
41 %
42 % FIX 6 - Locus-slice index clipping
. The original phN(1,k-r:r+k) can
43 %     go out of bounds when k <
r+1 or k+r > numel(w). The indices
44 %     are now clamped:

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4 %the linear model of the aircraft sys
tem with the pilot output as the in
put and theta as the output
5 %to include any feedback gains. The
linear model should be named 'namelin
.mdl'. Once the pilot
6 %model
7 %is chosen the second model is devel
oped called 'nameolop.mdl'. It has
the pilot model in the loop
8 %and has the output of the rate limit
er as its input and the output is th
e input into the rate limiter.

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45 %             iLo = max(1,      k
- r)
46 %             iHi = min(numel(w), k
+ r)
47 %
48 %   FIX 7 - Marker-style ladder replac
ed with cell arrays indexed by n
49 %             (markerStyles, markerFaces
). Identical visual result, far
50 %             shorter source, and trivia
l to extend past 7 rate-limit cases.
51 %
52 %   FIX 8 - Cosmetic: xlabel / ylabel
/ title on figure(4); consistent
53 %             section banners and indent
ation. No numerical impact.
54 %
55 %=====
=====
=====
56 %   Analytical relations used in FIX 1
:
57 %
58 %       Yc      = theta / pilot_stic
k      = K * Gc * Gac / (1 + Gc * Gac
)
59 %       P_RLE = RLE_in / pilot_stic
k      = K * Gc      / (1 + Gc * Gac)
60 %       OLOP  = Pilot * Yc
61 %=====
=====
=====
62
63 clear; clc; close all
64 format short
65
66 % =====
=====
=====
67 %   FIX 2: Aircraft model (AIAA-95-320
4-CP) - inlined from lin_abcd.m
68 % =====
=====
=====
69 %   Loop topology (rate limiter show
n for context, identity in linear mo
del)
70 %
71 %       qco --> [ K ] --(+)--> [ Gc
] --> delta_c --[ RLE ]--> delta

```

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9 %*****
*****
*
10 clear
11 format short
12 lin_abcd %Initialization File to ge
t aircraft model

```

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72 %          ^
73 %          |
74 %          v
75 %          '-----'
76 % Forward-loop gain
77 K = 13.68;
78
79 % Controller (pilot error -> RLE input)
80 Gc = tf( 5.21 * conv([1 -57.36], conv(
      ([1 4.26], [1 0.55])), conv([1 2*0.44
      2*22.85 22.85^2], conv([1 0], [1 1.16
      ])) ));
81
82 % Actuator dynamics (RLE output -> theta)
83 Gac = tf(-10.524 * conv([1 1.562], conv(
      [1 0.038], [1 0])), conv([1 2*0.2
      12*0.088 0.088^2], conv([1 3.75], [1
      -1.44])) );
84
85 % =====
86 % OLOP analysis
87 % =====
88
89 w=logspace(-2,2,1000);
90
91 % ----- User input
92 s -----
93 -----
94
95 R=input(' Enter Vector of Rate Limits to evaluate [default = 60]: ');
96 if isempty(R), R = 60; end
97
98 Per = input(' Enter Stick Input Amplitude in percentage (range 0-100) [default = 100%]: ');
99 if isempty(Per), Per = 100; end
100
101 Po = Per/100 * 3.6;
102 if Po < 0.
103     % non-linear stick gradient
104     Po = 2.5 * Po;
105 else

```

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13 w=logspace(-2,2,1000);

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14 %Define the rate limiter value and the pilot input amplitude

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15 R=input(' Enter Vector of Rate Limits to evaluate [default = 60]: ');
16 if isempty(R)

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102	Po = 2.5 * (2.86*Po - 1.48);	17	R=60;
103	end	18	end
104		19	
105	tag = input(' Choose desired pilot model (1=Low 2=Med 3=High 4=Neal-Smith) [default Neal-Smith]: ');	20	Per=input(' Enter Stick Input Amplitude in percentage (range 0-100) [default = 100%]: ');
106	if isempty(tag)	21	if isempty(Per)
107	tag = 4;	22	Per=100;
108	elseif tag == 1, xover = -90;	23	end
109	elseif tag == 2, xover = -110;	24	
110	elseif tag == 3, xover = -130;	25	Po=Per/100*3.6
111	end	26	if Po<.8 %Takes into account non-linear stick gradient
112		27	Po=2.5*Po;
113	% ----- Pilot model parameters (Neal-Smith) -----	28	else Po=2.5*(2.86*Po-1.48);end
114	Kp = -0.05 * 2.5;	29	
115	Tp1 = 0.06;	30	tag=input(' Choose desired pilot model (1=Low Gain 2=Med Gain 3=High gain 4=Neal-Smith) [default Neal-Smith]: ');
116	Tp2 = 0.01;	31	if isempty(tag)
117		32	tag=4;
118	% ----- Controlled element (theta / pilot_stick, closed loop) -----	33	elseif tag==1; xover=-90; %Set crossover angle for gain pilot model
119	% FIX 1: was [A,B,C,D] = LINMOD('nt33v2Dlin'); Yc = ss(A,B,C(4,:),D(4,:));	34	elseif tag==2; xover=-110;
120	Yc = ss(K*Gc*Gac / (1 + Gc*Gac));	35	elseif tag==3; xover=-130;
121		36	end
122	% ----- Pure-gain pilot search -----	37	
123	% FIX 4: silence the "closed-loop unstable" margin warnings during the sweep	38	Kp=-.05*2.5; Tp1=.06; Tp2=.01; %Gain, lead, and lag time constants for Neal-Smith Pilot Model
124	warning('off', 'Control:analysis:MarginUnstable')	39	%Use LINMOD to find appropriate transfer function for Yc=theta/pilot
125		40	[A,B,C,D]=LINMOD('nt33v2Dlin');
		41	Yc=ss(A,B,C(4,:),D(4,:)); %Make sure appropriate row is selected for theta output
		42	

<pre> 126 Kg=[(0.1*2.5):0.01:10; 127 for i=1:length(Kg) 128 [gm,pm,wcg,wcp]=margin(Kg(i)*Yc); 129 if tag==4, break, end 130 if -180+pm > xover 131 xover_angle=-180+pm 132 Pilot_gain=Kg(i) 133 break 134 end 135 end 136 137 warning('on', 'Control:analysis:MarginUnstable') 138 139 figure(1), ngridneal, nichols(Kg(i)*Yc, w) 140 grid on 141 % ----- Neal-Smith modified pilot model 142 bw=3.5; % bandwidth requirement 143 Yp=Kp*tf([5 1],[1 0])*tf([Tp1 1],[Tp2 1]); 144 set(Yp, 'InputDelay', 0.25); 145 Ypp=pade(Yp,2); % Pade approximation for ss/series ops 146 147 figure(2), ngridneal, nichols(Yp*Yc, {0.1, 3*bw}) % FIX 3: ngridneal is local 148 grid on 149 % ----- Closed-loop RLE input / pilot stick 150 % FIX 1: was P_RLE=ss(A,B,C(6,:),D(6,:)); 151 P_RLE = ss(K*Gc / (1 + Gc*Gac)); 152 [magP, phP]=bode(P_RLE,w); 153 154 % ===== 155 % Loop over rate limits </pre>	<pre> 43 %Find pure gain pilot model 44 Kg=.1*2.5:.01:10; 45 for i=1:length(Kg) 46 [gm,pm,wcg,wcp]=margin(Kg(i)*Yc); 47 if tag==4; break, end 48 if -180+pm > xover 49 xover_angle=-180+pm; 50 Pilot_gain=Kg(i) 51 break 52 end 53 end 54 55 figure(1),ngridneal,nichols(Kg(i)*Yc,w) 56 57 %Find Neal-Smith Modified Pilot model 58 bw=3.5;%Set bandwidth requirement 59 Yp=Kp*tf([5 1],[1 0])*tf([Tp1 1],[Tp2 1]); set(Yp,'InputDelay',.25); 60 Ypp=pade(Yp,2);%Need pade approximation for linmod to get OLOP transfer function 61 figure(2),ngridneal,nichols(Yp*Yc,{.1,3*bw}) 62 63 %Find closed-loop response amplitude from stick input to RLE input 64 P_RLE=ss(A,B,C(6,:),D(6,:)); 65 [magP,phP]=bode(P_RLE,w); 66 </pre>
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<pre> 156 % ===== 157 for n=1:length(R) 158 159 % Rate-limit line for the frequency plot: R / (j*omega) 160 [magR, phR] = bode(tf(R(n), [1 0]), w); 161 162 figure(3) 163 164 semilogx(w, 20*log10(Po*magP(1,:)), w, 20*log10(magR(1,:)), 'k:') 165 hold on, axis([0.01, 10, 10, 60]) 166 title('Determine Closed-loop Onset Frequency') 167 168 % Closed-loop onset frequency: Po * P_RLE(jw) meets the rate-limit line 169 iter = 100; 170 for i=1:length(w) 171 dif(i) = abs(magR(i) - Po*magP(i)); 172 if dif(i) <= iter 173 iter = dif(i); 174 k = i; 175 end 176 end 177 w_onset = w(k) 178 Rate = R(n) 179 180 % ----- OLOP transfer function ----- 181 % FIX 1: was [AA,BB,CC,DD] = linmod('nt33v2dolop'); 182 % OLOP = ss(AA,BB,-CC,-DD); 183 if tag==4 184 Pilot = Ypp; % Neal-Smith with Pade-approx delay 185 else 186 Pilot = tf(Pilot_gain, 1); % pure-gain pilot 187 end </pre>	<pre> 67 for n=1:length(R) 68 69 %Calculate Rate Limit Line for frequency plot 70 [magR, phR] = bode(tf(R(n), [1 0]), w); 71 72 %Plot closed-loop amplitude and rate limit line 73 figure(3), semilogx(w, 20*log10(Po*magP(1,:)), w, 20*log10(magR(1,:)), 'k:') 74 hold on, axis([.01, 10, 10, 60]) 75 title('Determine Closed-loop Onset Frequency') 76 77 %Calculate omega onset for closed-loop system 78 iter=100; 79 for i=1:length(w) 80 dif(i) = abs(magR(i) - Po*magP(i)); 81 if dif(i) <= iter 82 iter = dif(i); k=i; 83 end 84 end 85 w_onset=w(k), Rate=R(n) 86 87 %Calculate OLOP on Nichols chart 88 89 if tag==4 90 Pilot=Ypp; %Neal-Smith Pilot Model with Pade approximation for time delay 91 else 92 Pilot=tf(Pilot_gain,1); %Simple Gain Pilot Model 93 end </pre>
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<pre> 187 OLOP = Pilot * Yc ; % pilot * control led element 188 189 % FIX 5: squeeze 3-D nichols output to row vectors so 1-D indexing works 190 [magN, phN] = nichols(OLOP, w); 191 magN = squeeze(magN).'; 192 phN = squeeze(phN).'; 193 194 % ----- OLOP point + locus on the Nichols chart ----- 195 r = 150 ; % plot range around onset index 196 v1 = [-60 -90 -100 -120 -140 -160 -180]; 197 v2 = [13.5 7.5 5.5 2.5 1.1 0 0]; ; % OLOP stability boundary 198 199 % FIX 7: marker ladder collapse into indexed cell arrays 200 markerStyles = {'o', 'd', 'o', 'p', 's', 'd', 's'}; 201 markerFaces = {'k', 'k', 'none', 'k', 'k', 'none', 'none'}; 202 idx = min(n, numel(markerStyles)); 203 204 figure(4) 205 plot(phN(k), 20*log10(magN(k)), markerStyles{idx}, 'MarkerFaceColor', markerFaces{idx}, 'MarkerSize', 7); 206 grid on 207 hold on 208 209 % FIX 6: clamp the locus slice so it never indexes out of bounds </pre>	<pre> 91 92 [AA, BB, CC, DD]=linmod('nt33v2dolop'); 93 OLOP=ss(AA, BB, -CC, -DD); 94 [magN, phN]=nichols(OLOP, w); 95 r=150; %Plot range 96 v1=[-60 -90 -100 -120 -140 -160 -180]; v2=[13.5 7.5 5.5 2.5 1.1 0 0]; %OLOP stability boundary 97 %Plot OLOP points for each rate value 98 if n==1 99 figure(4), plot(phN(k), 20*log10(magN(k)), 'o', 'MarkerFaceColor', 'k', 'MarkerSize', 7) 100 elseif n==2 101 figure(4), plot(phN(k), 20*log10(magN(k)), 'd', 'MarkerFaceColor', 'k', 'MarkerSize', 7) 102 elseif n==3 103 figure(4), plot(phN(k), 20*log10(magN(k)), 'o', 'MarkerSize', 7) 104 elseif n==4 105 figure(4), plot(phN(k), 20*log10(magN(k)), 'p', 'MarkerFaceColor', 'k', 'MarkerSize', 7) 106 elseif n==5 107 figure(4), plot(phN(k), 20*log10(magN(k)), 's', 'MarkerFaceColor', 'k', 'MarkerSize', 7) 108 elseif n==6 </pre>
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<pre> 210 iLo = max(1, k - r); 211 iHi = min(numel(w), k + r); 212 plot(phN(iLo:iHi), 20*log10(magN # (iLo:iHi)), [-360 0], [0 0], '-k', [-180 -180], [-50 50], '-k') 213 214 axis([-200, -60, -10, 20]); 215 plot(v1, v2, '-k', 'LineWidth' # , 1.5) % OLOP stability boundary 216 grid on 217 end 218 219 % FIX 8: figure-4 labels 220 xlabel('Open-loop phase [deg]') 221 ylabel('Open-loop magnitude [dB]') 222 title('OLOP Nichols chart with stabil ity boundary') 223 224 225 %% ===== ===== == 226 % FIX 3: ngridneal embedded as a loc al function (was a missing helper) 227 % ===== ===== == 228 229 function ngridneal 230 %NGRIDNEAL Nichols grid for Neal-Smi th / OLOP pilot-vehicle analysis. 231 % 232 % Stand-in for the (missing) ngridn eal helper from the original 233 % Gilbreath OLOP package. Draws: 234 % (1) the standard Nichols grid (closed-loop M and N contours) 235 % (2) a +3 dB M-contour highlight ed in red - the Neal-Smith 236 % Level-1 / Level-2 closed-lo op resonant-peak boundary used in 237 % pilot handling-qualities an alysis. 238 239 ngrid 240 241 hold on # 242 ax = gca; 243 244 % Highlight the +3 dB M-circle. 245 % Parametric trace of T = M in the L = T/(1-T) plane: </pre>	<pre> 109 figure(4), plot(phN(k), 20*log10(magN(k)), 'd', 'MarkerSize', 7) 110 else 111 figure(4), plot(phN(k), 20*log10(magN(k)), 's', 'MarkerSize', 7) 112 end 113 hold on </pre>
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<pre> 246 M_dB = 3; 247 M = 10^(M_dB/20); 248 phi = linspace(eps, 2*pi - eps, 249 720); 250 T = M * exp(1i*phi); 251 252 magL_dB = 20*log10(abs(L)); 253 phaseL = unwrap(angle(L)) * 180/ pi; 254 phaseL = mod(phaseL, 360) - 360 ; % wrap to [-360, 0] 255 256 plot(ax, phaseL, magL_dB, 'r-' , 'LineWidth', 1.4, 'HandleVisibilit y', 'off'); 257 text(ax, -180, M_dB + 1, sprintf(' M = %g dB (NS bound)', M_dB), 'Co lor', 'r', 'FontSize', 8, 'HandleVisi bility', 'off'); 258 grid on 259 260 hold off 261 end </pre>	<pre> + 114 %Plot mag and phase on Nichols chart 115 plot(phN(1,k-r:r+k),20*log10(magN(1, k-r:r+k)),... 116 [-360 0],[0 0],'-k',[-180 -180],[-50 50],'-k') 117 axis([-200,-60,-10,20]); 118 plot(v1,v2,'-k','LineWidth',1.5)%Plo t OLOP stability boundary 119 end </pre>
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